1. Project Title

Citizen Safety and Emergency Coordination Platform with Mobile Application

2. Project Statement and Outcomes

Senior citizens, children, and dependent individuals often spend time alone at home while family members are away for work, travel, business commitments, or other personal engagements. During emergencies such as health issues, accidents, falls, security concerns, or situations requiring immediate assistance, guardians may not always be reachable.

In residential communities, gated societies, and apartment complexes, there are often neighbors, volunteers, and security personnel available nearby who can provide immediate assistance. However, there is currently no structured platform that enables rapid communication, emergency escalation, and coordinated community response.

This project aims to develop a Community Emergency Response & Assistance Network consisting of a web-based administration portal and a mobile application that allows residents to trigger emergency assistance requests instantly. Upon SOS activation, notifications are sent to designated guardians, security personnel, volunteers, and nearby society members based on configurable escalation workflows.

The platform will provide real-time communication, incident

tracking, emergency notifications, guardian escalation, and community response coordination to ensure timely assistance during emergencies.

The deliverable is a complete Python-based solution comprising a backend platform, administrative web portal, Android/iOS mobile application, notification services, emergency management workflows, reporting dashboards, and deployment infrastructure.

3. Modules to be Implemented

1. Resident, Society & Emergency Contact Management System

2. SOS Alert & Emergency Notification Management System

3. Community Response & Guardian Escalation Platform

4. Mobile Application Development

5. System Integration, Testing & Project Finalization

4. Week-wise Module Implementation and High-Level Requirements

Weeks 1–2: Resident, Society & Emergency Contact Management System

User Management

- Resident Registration

- Guardian Registration
- Volunteer Registration
- Security Personnel Registration
- User Authentication & Authorization

Society Management

- Society Registration
- Block / Tower Management
- Flat Management
- Resident Mapping

Emergency Contact Management

- Primary Guardian Configuration
- Secondary Guardian Configuration
- Emergency Contact Configuration
- Contact Verification

Backend Development

- Django REST Framework project setup
- PostgreSQL database design
- REST API development
- Role-based access control

Weeks 3–4: SOS Alert & Emergency Notification Management System

SOS Management

- One-Tap SOS Activation
- Emergency Category Selection
- Emergency Message Creation
- Location Capture

Notification Engine

- Push Notifications
- SMS Notifications
- Email Notifications
- In-App Notifications

Alert Routing

- Primary Guardian Notification
- Security Personnel Notification
- Volunteer Notification
- Community Broadcast Notification

Alert Monitoring

- Alert Status Tracking
- Notification Delivery Tracking

- Response Monitoring

Weeks 5–6: Community Response & Guardian Escalation Platform

Guardian Escalation Workflow

- Notify Primary Guardian
- Configurable Response Time Window
- Escalate to Secondary Guardian
- Escalate to Emergency Contacts

Community Assistance Management

- Volunteer Availability Status
- Incident Acceptance Workflow
- Responder Assignment
- Assistance Tracking

Communication Services

- Emergency Chat Module
- Contact Directory
- Response Updates
- Incident Discussion Thread

Incident Lifecycle Management

- Open Incident
- Active Response
- Escalated Incident
- Resolved Incident
- Closure Documentation

Weeks 7–8: Mobile Application Development, Testing & Deployment

Mobile Application Development

Resident Mobile App

- SOS Button
- Emergency Contact Management
- Incident Tracking
- Notification Center
- Profile Management

Guardian Mobile App Features

- Real-Time SOS Alerts
- Incident Monitoring
- Emergency Response Tracking
- Escalation Notifications

Volunteer Mobile App Features

- Incident Notifications
- Response Acceptance
- Assistance Updates
- Availability Management

Security Staff Features

- Incident Dashboard
- Response Coordination
- Incident Resolution Updates

Testing & Optimization

- API Testing
- Mobile Application Testing
- Security Testing
- Performance Testing
- User Acceptance Testing

Documentation

- System Architecture
- Database Design
- API Documentation
- Mobile App Documentation
- User Manual

Final Presentation

- Problem Statement
- Solution Overview
- Architecture Diagram
- Mobile App Demonstration
- Workflow Demonstration
- Social Impact Assessment

5. Flow Chart

Resident Initiates SOS

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SOS Request Created

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Primary Guardian Notified

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Security Team Notified

↓

Volunteers Notified

↓

Community Members Alerted

↓

Response Received?

↓ Yes

Responder Assigned

↓

Assistance Provided

↓

Incident Closed

↓ No

Escalate to Secondary Guardian

↓

Escalate to Emergency Contacts

↓

Incident Resolution

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Reporting & Analytics